Expt No : 12a
Date : 8/10/26

ECHO CLIENT SERVER using TCP and UDP Sockets

AIM:

To implement an Echo client Server using TCP and UDP Sockets

1. TCP Echo Server -

```
import socket
Server = Socket.socket(Socket.AF_INET, Socket.SOCK_STREAM)
Server.bind(("localhost", 12345))
Server.listen(1)
Print("TCP Echo Server is waiting for a client...")
Conn, addr = Server.accept()
Print("Connected to :", addr)
While True:
    data = Conn.recv(1024)
    if not data:
        break
    Print("Client :", data.decode())
    Conn.sendall(data)
Conn.close()
Server.close()
```

2. TCP Echo client :

```
import socket
Client = Socket.socket(Socket.AF_INET, Socket.SOCK_STREAM)
Client.connect(("localhost", 12345))
While True:
    message = input("Enter message (Exit to stop): ")
    if message.lower() == "exit":
        break
    Client.sendall(message.encode())
    data = Client.recv(1024)
    Print("Echo from Server :", data.decode())
Client.close()
```

3. UDP Echo Server:

```
import socket
server = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
server.bind("localhost", 12346))
Print("UDP Echo Server is waiting for messages...")
while True:
    data, addr = server.recvfrom(1024)
    Print("Client:", data.decode())
    if data.decode().lower() == "exit":
        server.sendto(data, addr)
        Continue
    server.sendto(data, addr)
```

4. UDP Echo Client:

```
import socket
client = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
while True:
    message = input("Enter message (exit to stop): ")
    client.sendto(message.encode(), ("localhost", 12346))
    data, addr = client.recvfrom(1024)
    Print("Echo from server: ", data.decode())
    if message.lower() = "exit":
        break
client.close()
```

```
PS C:\Users\admin\Downloads\EXP\EXP12> py tcp_server.py
TCP Echo Server is waiting for a client...
Connected to: ('127.0.0.1', 50380)
Client: HELLO
Client: HI
PS C:\Users\admin\Downloads\EXP\EXP12> py udp_server.py
UDP Echo Server is waiting for messages...
Client: GOOD
Client: AWESOME
Client: exit
```

```
PS C:\Users\admin\Downloads\EXP\EXP12> py tcp_client.py
Enter message (exit to stop): HELLO
Echo from server: HELLO
Enter message (exit to stop): HI
Echo from server: HI
Enter message (exit to stop): exit
PS C:\Users\admin\Downloads\EXP\EXP12> py udp_client.py
Enter message (exit to stop): GOOD
Echo from server: GOOD
Enter message (exit to stop): AWESOME
Echo from server: AWESOME
Enter message (exit to stop): exit
Echo from server: exit
PS C:\Users\admin\Downloads\EXP\EXP12>
```

RESULT:

Thus the Echo client Server is done executed Successfully using TCP and UDP Packets